# Problems

## 1H–2

Using Cramer's rule, give another proof that if $A$ is an $n \times n$ matrix whose determinant is non-zero, then the equation

$$AX = 0$$

has only the trivial solution.

## 1H–3

**(a)** For what values of $c$ will the system

$$\begin{cases} x_1 - 2x_2 + cx_3 = 0 \\ 2x_1 + x_2 + x_3 = 0 \\ -x_1 + 2x_3 = 0 \end{cases}$$

have a non-trivial solution?

**(b)** For what values of $c$ will

$$\begin{pmatrix} 2 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = c \begin{pmatrix} x \\ y \end{pmatrix}$$

Write it as a homogeneous system.

**(c)** For each value of $c$ in part (a), find a non-trivial solution to the corresponding system.

**(d)** For each value of $c$ in part (b), find a non-trivial solution to the corresponding system.

## 1H–4

Find all solutions to the homogeneous system

$$\begin{cases} x - 2y + z = 0 \\ x + y - z = 0 \\ 3x - 3y + 2z = 0 \end{cases}$$

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# Assignment

## 1H–8 (continued)

Suppose

$$\frac{ax+b}{(x-r_1)(x-r_2)} = \frac{c}{x-r_1} + \frac{d}{x-r_2},$$

where $a, b, r_1, r_2$ are given and $c, d$ are to be found.

What are the linear equations which determine the constants $c$ and $d$? Under what circumstances do they have a unique solution?

# II. Vector Functions and Parametric Equations

## II–1

The point $P$ moves with constant speed $v$ in the direction of the constant vector

$$a\mathbf{i} + b\mathbf{j}.$$

If at time $t = 0$ it is at $(x_0, y_0)$, what is its position at time $t$? What is its position vector function $\mathbf{r}(t)$?

## II–2

A point moves clockwise with constant angular velocity $\omega$ on the circle of radius $a$ centered at the origin. What is its position vector function $\mathbf{r}(t)$ if, at time $t = 0$, it is at

(a) $(a, 0)$ (b) $(0, a)$?

## II–3

Describe the motions given by each of the following position vector functions as $t$ goes from $-\infty$ to $\infty$. In each case, give the $xy$-equation of the curve along which $P$ travels and tell what part of the curve is actually traced out by $P$.

(a) $\mathbf{r} = 2\cos^2 t\,\mathbf{i} + \sin^2 t\,\mathbf{j}$

(b) $\mathbf{r} = \cos 2t\, \mathbf{i} + \cos t\, \mathbf{j}$

(c) $\mathbf{r} = (t^2 + 1)\mathbf{i} + t^3\mathbf{j}$

(d) $\mathbf{r} = \tan t\, \mathbf{i} + \sec t\, \mathbf{j}$

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## II-6

A bow-and-arrow hunter walks toward the origin along the positive $x$-axis with unit speed. At time $t = 0$ he is at $x = 10$. His arrow (of unit length) is aimed toward a rabbit hopping with constant velocity. It is in the first quadrant along the line $y = 2x$; at time $t = 0$ it is at the origin.

(a) Write down the vector function $\vec{A}(t)$ for the arrow at time $t$.

(b) The hunter shoots (and misses). When is the arrow closest to the rabbit?

## II-7

The cycloid is the curve traced out by a fixed point $P$ on a circle of radius $a$ which rolls along the $x$-axis in the positive direction, starting when $P$ is at the origin $O$.

Find the vector function $\vec{OP}$, using as variable the angle $\theta$ through which the circle has rolled.

**Hint:** Begin by expressing $\vec{OP}$ as the sum of three simpler vector functions.

# Vectors and Matrices

## Differentiation of Vector Functions

**IJ-1** For each of the vector functions of time, calculate the velocity, speed $\left|\frac{ds}{dt}\right|$, unit tangent vector (in the direction of velocity), and acceleration.

(a) $\mathbf{r}(t) = e^t\mathbf{i} + te^t\mathbf{j}$

(b) $\mathbf{r}(t) = 2\mathbf{i} + t^3\mathbf{j}$

(c) $\mathbf{r}(t) = (1 - 2t + t^2)\mathbf{i} + (-2 + 2t^2)\mathbf{k}$

**IJ-2** Let

$$\vec{OP} = \frac{1}{1+t^2}\mathbf{i} + \frac{t}{1+t^2}\mathbf{j}$$

be the position vector for a motion.

(a) Calculate $\mathbf{v}$, $\left|\frac{ds}{dt}\right|$, and $\mathbf{T}$.

(b) At what point is the speed greatest? Smallest?

(c) Find the $xy$-equation of the curve along which the point $P$ is moving and describe it geometrically.

**IJ-3** Prove the rule for differentiating the scalar product of two plane vector functions:

$$\frac{d}{dt}(\mathbf{r}\cdot\mathbf{s}) = \frac{d\mathbf{r}}{dt}\cdot\mathbf{s} + \mathbf{r}\cdot\frac{d\mathbf{s}}{dt}$$

by calculating with components, letting

$$\mathbf{r} = x_1\mathbf{i} + y_1\mathbf{j}, \qquad \mathbf{s} = x_2\mathbf{i} + y_2\mathbf{j}.$$